PRILL-EXCEPTIONAL ÉTALE COVERS FROM RAYNAUD BUNDLES

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Abstract. Let C be a smooth projective connected complex curve of genus $g \geq 2$. We prove that C admits a connected finite étale Galois cover $f\colon X \to C$ such that

$$h^0\Big(X, O_X\big(f^{-1}(p)\big)\Big) \geq 2$$

for every $p \in C$. Let $V = R_g \otimes M$ be a determinant-normalised twist of the rank-g^g Raynaud bundle. Then $\deg V = 0$, $\det V \cong O_C$, and $H^0(C, V) = 0$, whereas $H^0(C, V(p)) \neq 0$ for every $p \in C$. We prove that V has finite monodromy and choose an irreducible summand W with $H^0(C, W(p)) \neq 0$ for all p. For the Galois cover defined by the monodromy of W, the associated bundle $f_* O_X$ contains O_C and W as direct summands, which yields the claimed inequality.

1. Introduction

Let $f\colon X \to C$ be a finite morphism of smooth projective connected complex curves. Prill asked whether a fibre divisor $f^{-1}(p)$ can move in a pencil; equivalently, whether

$$h^0\Big(X, O_X\big(f^{-1}(p)\big)\Big) > 1$$

for a general point $p \in C$. The question appears in [1, Chapter VI, Exercise D] and, for curves of genus at least three, is recorded as [6, Problem 14]. Partial affirmative results were obtained in [3, 7]. Landesman and Litt subsequently constructed, for every curve of genus two, a connected finite étale cover of degree 36 satisfying the inequality for every $p \in C$ [5].

We call f *Prill-exceptional* when the inequality holds for every $p \in C$. The construction below works for every $g \geq 2$.

Theorem 1.1. *Let C be a smooth projective connected complex curve of genus $g \geq 2$. There is a connected finite étale Galois cover $f\colon X \to C$ such that*

$$h^0\Big(X, O_X\big(f^{-1}(p)\big)\Big) \geq 2$$

for every $p \in C$.

For $g \geq 3$, Theorem 1.1 answers [6, Problem 14] affirmatively. For a finite flat morphism, the function

$$p \longmapsto h^0\Big(X, O_X\big(f^{-1}(p)\big)\Big)$$

is upper semicontinuous. The locus where this dimension is at least two is therefore closed. If it contains a dense open subset of C, it equals C. Thus the general-fibre and pointwise formulations are equivalent. We use the latter because the construction yields it directly.

The proof uses two properties of Raynaud's construction. The restricted bundle R_g satisfies $H^0(C, R_g \otimes \alpha) \neq 0$ for every $\alpha \in \mathrm{Pic}^0(C)$, whereas the ambient bundle E_g becomes a direct sum of copies of one line bundle after pullback by $[g]\colon J \to J$. We choose a twist with trivial determinant while preserving the pointwise nonvanishing. On the induced étale pullback of C, the repeated line bundle is torsion, so the twisted bundle has finite linear monodromy.

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2. The Raynaud bundle

Write $J = \mathrm{Jac}(C) = \mathrm{Pic}^0(C)$, fix a theta divisor Θ on J, and choose an Abel–Jacobi embedding $i\colon C \hookrightarrow J$. The form of Raynaud's construction needed here is recorded in [8, Section 3] and [2, Section 2.3].

Theorem 2.1 (Raynaud)**.** *For every integer $n \geq 1$, there is a vector bundle E_n on J such that*

$$[n]^* E_n \cong H^0(J, \mathcal{O}_J(n\Theta))^* \otimes_{\mathbb{C}} \mathcal{O}_J(n\Theta). \tag{1}$$

Its restriction $R_n = i^* E_n$ is semistable, has rank n^g and slope g/n, and satisfies

$$H^0(C, R_n \otimes \alpha) \neq 0 \qquad \textit{for every } \alpha \in \mathrm{Pic}^0(C). \tag{2}$$

Take $n = g$ and set

$$R := R_g, \qquad r := \mathrm{rk}(R) = g^g.$$

Then R is semistable of slope one, so $\deg(R) = r$.

3. A determinant-normalised twist

Lemma 3.1. *There is a line bundle $M \in \mathrm{Pic}^{-1}(C)$ such that*

$$M^{\otimes r} \cong (\det R)^{-1} \qquad \textit{and} \qquad H^0(C, R \otimes M) = 0. \tag{3}$$

Proof. The map

$$\mathrm{Pic}^{-1}(C) \longrightarrow \mathrm{Pic}^{-r}(C), \qquad M \longmapsto M^{\otimes r},$$

is a translate of the multiplication-by-r isogeny of J. Consequently, the fibre

$$T := \left\{ M \in \mathrm{Pic}^{-1}(C) : M^{\otimes r} \cong (\det R)^{-1} \right\}$$

has cardinality r^{2g}.

Call $M \in \mathrm{Pic}^{-1}(C)$ *bad* if $H^0(C, R \otimes M) \neq 0$. Such a section induces a nonzero morphism $M^{-1} \to R$. If $L \subset R$ is the saturation of its image, then $L \cong M^{-1}(D)$ for an effective divisor D. Since $\deg(M^{-1}) = 1$ and R is semistable of slope one,

$$1 + \deg D = \deg L \leq 1.$$

Thus $D = 0$, and M^{-1} is a degree-one line subbundle of R.

Fix a Jordan–Hölder filtration $0 = R_0 \subset R_1 \subset \cdots \subset R_m = R$. Given a degree-one line subbundle $L \subset R$, let k be minimal with $L \subset R_k$. The induced map $L \to R_k/R_{k-1}$ is nonzero. Since its source and target are stable of slope one, it is an isomorphism. Thus every bad M makes M^{-1} isomorphic to a rank-one Jordan–Hölder factor of R. The filtration has at most r factors, so at most r isomorphism classes of M are bad.

Because $g \geq 2$ and $r = g^g > 1$, one has $r^{2g} > r$. The set T therefore contains an M that is not bad. □

Fix M as in Lemma 3.1 and put $V := R \otimes M$. Then

$$\deg V = 0, \qquad \det V \cong \mathcal{O}_C, \qquad H^0(C, V) = 0. \tag{4}$$

For every $p \in C$, the line bundle $M(p)$ has degree zero, so (2) gives

$$H^0(C, V(p)) = H^0(C, R \otimes M(p)) \neq 0. \tag{5}$$

4. Finite monodromy

Proposition 4.1. *The vector bundle V is trivialised by a connected finite étale cover of C.*

Proof. Form the cartesian square

$$\begin{tikzcd} \widetilde{C} \arrow[r, "\widetilde{i}"] \arrow[d, "q"'] & J \arrow[d, "{[g]}"] \\ C \arrow[r, "i"] & J. \end{tikzcd}$$

The morphism q is finite étale and surjective. By (1),

$$q^*V \cong H^0(J, O_J(g\Theta))^* \otimes_{\mathbb{C}} \left(\widetilde{i}^* O_J(g\Theta) \otimes q^*M\right).$$

Let $\widetilde{C}_0$ be a connected component of $\widetilde{C}$. Its image under q is nonempty, open, and closed, hence equals C. Define

$$N := \left(\widetilde{i}^* O_J(g\Theta) \otimes q^*M\right)\Big|_{\widetilde{C}_0}.$$

Since $h^0(J, O_J(g\Theta)) = g^g = r$, the restriction of q^*V to $\widetilde{C}_0$ is $N^{\oplus r}$. Taking determinants and using $\det V \cong O_C$ yields

$$N^{\otimes r} \cong O_{\widetilde{C}_0}.$$

Let d be the order of N; then $d \mid r$. A chosen isomorphism $N^{\otimes d} \cong O_{\widetilde{C}_0}$ defines the finite étale μ_d-torsor

$$\mathrm{Spec}_{\widetilde{C}_0}\left(\bigoplus_{a=0}^{d-1} N^{-a}\right) \longrightarrow \widetilde{C}_0.$$

The pullback of N to this torsor is trivial. Any connected component maps surjectively to $\widetilde{C}_0$; its composite with $\widetilde{C}_0 \to C$ is a connected finite étale cover $h\colon C' \to C$ that trivialises V. □

Passing to a connected finite étale Galois cover dominating h, we may regard V as the bundle associated with a complex representation of a finite group.

Proposition 4.2. *There is a nontrivial irreducible finite-monodromy vector bundle W on C such that*

$$H^0(C, W(p)) \neq 0 \qquad \textit{for every } p \in C.$$

Proof. Choose a connected finite étale Galois cover $u\colon Y \to C$ trivialising V, and write H for its Galois group. Fix a trivialisation $u^*V \cong O_Y^{\oplus r}$. Since Y is connected and projective, $H^0(Y, O_Y) = \mathbb{C}$; hence the descent matrices are constant and define a representation $H \to \mathrm{GL}_r(\mathbb{C})$. Maschke's theorem gives a decomposition

$$V \cong \bigoplus_{j=1}^{s} W_j^{\oplus m_j},$$

where the W_j are pairwise nonisomorphic irreducible finite-monodromy bundles. No W_j is trivial, because a trivial summand would contradict $H^0(C, V) = 0$ in (4).

For $1 \leq j \leq s$, set

$$Z_j := \{p \in C \mid H^0(C, W_j(p)) \neq 0\}.$$

Let $p_1, p_2\colon C \times C \to C$ be the projections and let $\Delta \subset C \times C$ be the diagonal. The restriction of $p_1^*W_j \otimes O_{C\times C}(\Delta)$ to $C \times \{p\}$ is $W_j(p)$. Upper semicontinuity for the proper morphism p_2 shows that each Z_j is closed. The decomposition of V and (5) give $C = \bigcup_{j=1}^{s} Z_j$. An irreducible variety cannot be a finite union of proper closed subsets, so $Z_j = C$ for some j. □

5. The covering

The vector-bundle criterion used below is the étale case of [4, Lemma 5.4]; we include the argument for completeness.

Proof of Theorem 1.1. Let W be as in Proposition 4.2, and let

$$\rho\colon \pi_1(C,c) \longrightarrow \mathrm{GL}(W_c)$$

be its monodromy representation. Its image G is finite. The subgroup $\ker\rho$ determines a connected finite étale Galois cover $f\colon X \to C$ with Galois group G.

The bundle $f_*\mathcal{O}_X$ corresponds to the regular G-module $\mathbb{C}[G]$, which contains every irreducible complex representation of G with multiplicity equal to its dimension. Hence $\mathcal{O}_C$ and W are distinct direct summands of $f_*\mathcal{O}_X$.

For $p \in C$, the fibre divisor is the pullback of p, so

$$\mathcal{O}_X\big(f^{-1}(p)\big) \cong f^*\mathcal{O}_C(p).$$

The projection formula and the summands $\mathcal{O}_C, W \subset f_*\mathcal{O}_X$ give

$$\begin{aligned} H^0\Big(X, \mathcal{O}_X\big(f^{-1}(p)\big)\Big) &\cong H^0(C, f_*\mathcal{O}_X \otimes \mathcal{O}_C(p)) \\ &\supset H^0(C, \mathcal{O}_C(p)) \oplus H^0(C, W(p)). \end{aligned}$$

Since p is effective, $h^0(C,\mathcal{O}_C(p)) \geq 1$; Proposition 4.2 gives $h^0(C, W(p)) \geq 1$. Thus the left-hand side has dimension at least two for every $p \in C$. □

Remark 5.1. The construction gives no useful degree bound. The initial Raynaud bundle has rank g^g, and the proof subsequently takes a Kummer cover and a Galois closure. In genus two, the degree-36 cover of [5] is numerically far more efficient.

Remark 5.2. The determinant condition is essential. Formula (1) shows only that the pullback of R_g is a direct sum of copies of one line bundle. Triviality of the determinant makes that line bundle torsion. Consequently, the twisted bundle has finite linear, rather than merely finite projective, monodromy.

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